

- 1) In Gauss elimination, back substitution is used after:
 - A. Forward elimination
 - B. Iteration
 - C. Differentiation
 - D. Integration
- 2) The Gauss-Seidel method is derived from:
 - A. Iteration formula
 - B. Elimination formula
 - C. Matrix inverse
 - D. Determinant
- 3) Numerical differentiation error depends on:
 - A. Step size
 - B. Matrix
 - C. Root
 - D. Area
- 4) Numerical differentiation is based on:
 - A. Finite differences
 - B. Infinite series
 - C. Integration
 - D. Matrix
- 5) The order of Simpson's 1/3 rule is:
 - A. 2
 - B. 3
 - C. 4
 - D. 5
- 6) The order of trapezoidal rule is:
 - A. 1
 - B. 2
 - C. 3
 - D. 4
- 7) Gauss-Seidel method is iterative because:
 - A. Uses repeated approximation
 - B. Uses elimination
 - C. Uses substitution
 - D. Uses matrix inverse
- 8) Numerical integration approximates:
 - A. Area under curve
 - B. Derivative
 - C. Root
 - D. Matrix
- 9) Gauss elimination is unsuitable when matrix is:
 - A. Singular
 - B. Diagonal
 - C. Identity
 - D. Symmetric
- 10) If the step size is halved in trapezoidal rule, the error approximately becomes:

- 11) A. Half
B. One-fourth
C. Double
D. One-eighth
- 12) If the step size is halved in Simpson's 1/3 rule, the error becomes approximately:
A. $1/2$
B. $1/4$
C. $1/8$
D. $1/16$
- 13) Simpson's 1/3 rule is exact for polynomials of degree:
A. ≤ 1
B. ≤ 2
C. ≤ 3
D. ≤ 4
- 14) Simpson's 3/8 rule is exact for polynomial of degree:
A. ≤ 1
B. ≤ 2
C. ≤ 3
D. ≤ 4
- 15) Trapezoidal rule is exact for polynomial of degree:
A. ≤ 1
B. ≤ 2
C. ≤ 3
D. ≤ 4
- 16) Composite Simpson's rule improves accuracy because:
A. Uses smaller intervals
B. Uses larger intervals
C. Uses elimination
D. Uses iteration
- 17) Gauss-Seidel method converges faster when matrix is:
A. Diagonally dominant
B. Singular
C. Symmetric
D. Rectangular
- 18) Gauss-Seidel method converges faster when matrix is:
A. Diagonally dominant
B. Singular
C. Symmetric
D. Rectangular
- 19) If diagonal dominance condition is not satisfied, Gauss-Seidel method may:
A. Converge slowly
B. Diverge
C. Give exact solution
D. Stop
- 20) Numerical differentiation becomes unstable when:
A. Step size very small
B. Step size large

- C. Function linear
 - D. Function constant
- 21) Central difference formula gives higher accuracy because:
- A. Uses surrounding points
 - B. Uses single point
 - C. Uses elimination
 - D. Uses integration
- 22) The truncation error of trapezoidal rule is proportional to:
- A. h
 - B. h^2
 - C. h^3
 - D. h^4
- 23) The truncation error of Simpson's 1/3 rule is proportional to:
- A. h^2
 - B. h^3
 - C. h^4
 - D. h^5
- 24) The truncation error of Simpson's 3/8 rule is proportional to:
- A. h^2
 - B. h^3
 - C. h^4
 - D. h^5
- 25) Numerical integration is more accurate when:
- A. Step size decreases
 - B. Step size increases
 - C. Step size constant
 - D. Step size zero
- 26) The Gauss elimination method is based on:
- A. Row transformation
 - B. Column transformation
 - C. Graph
 - D. Iteration
- 27) Pivot element in Gauss elimination should not be:
- A. Zero
 - B. One
 - C. Positive
 - D. Negative
- 28) Gauss-Seidel iteration converges if spectral radius is:
- A. < 1
 - B. > 1
 - C. $= 1$
 - D. ≥ 1
- 29) Simpson's 1/3 rule requires:
- A. Equal intervals
 - B. Unequal intervals
 - C. Random intervals
 - D. Variable intervals
- 30) Trapezoidal rule is a numerical method that approximates the value of:

- A. Improper integral
 - B. Indefinite integral
 - C. Definite integral
 - D. None of these
- 31) The aim of elimination steps in Gauss elimination method is to reduce the coefficient matrix to:
- A. Diagonal
 - B. Upper triangular
 - C. Lower triangular
 - D. Identity
- 32) Iteration is called as:
- A. Accurate process
 - B. Self-correcting process
 - C. Rounding-off process
 - D. None of the above
- 33) This technique is used to solve a system of linear first-order differential equations when the coefficient matrix is constant:
- A. Laplace Transform
 - B. Separation of Variables
 - C. Variation of Parameters
 - D. Matrix Exponentiation
- 34) The step size hhh in tabulated data is:
- A. Always one
 - B. The distance between consecutive y-values
 - C. The difference between x-values
 - D. Not required
- 35) The elimination process in Gauss elimination method is also known as:
- A. Crossway elimination
 - B. Sideway elimination
 - C. Forward elimination
 - D. Backward elimination
- 36) Gauss-Seidel method is similar to which of the following methods?
- A. Iteration method
 - B. Newton-Raphson method
 - C. Jacobi's method
 - D. Regula Falsi method
- 37) Which of the following method is employed for solving the system of linear equations?
- A. Runge-Kutta
 - B. Newton-Raphson
 - C. Gauss-Seidel
 - D. None of the above
- 38) Numerical techniques more commonly involve:
- A. Iterative method
 - B. Elimination method
 - C. Reduction method
 - D. Direct method
- 39) Which of the following is an iterative method?
- A. Gauss-Seidel
 - B. Gauss-Jordan
 - C. Factorization
 - D. Gauss elimination
- 40) Gauss-Seidel method is also termed as method of:
- A. Iterations
 - B. False position
 - C. Successive displacement
 - D. Eliminations

- 41) In which of the following both sides of the equation are multiplied by non-zero constants?
- Gauss elimination method
 - Gaussian inconsistent procedure
 - Gaussian substitute procedure
 - Gaussian consistent procedure
- 42) While solving by **Gauss-Seidel method**, which of the following is the first iterative solution for the system
 $x-2y=1$ and $x+4y=4$?
- (1, 0.75)
 - (0.25, 1)
 - (0, 0)
 - (1, 0.65)
- 43) Solve the following equations by **Gauss-Seidel method** up to first iteration and find the value of z:
 $27x+6y-z=72$, $6x+15Y+2z=72$, $x+y+54z=110$
- 1.82
 - 1.22
 - 0
 - 1.91
- 44) Numerical methods are used to:
- Solve exact solution analytically
 - Approximate solution to mathematical problem
 - Avoid using computers
 - Increase error in computations
- 45) Euler method approximates the solution of:
- Nonlinear algebraic equations
 - Ordinary differential equations
 - Integral equations
 - Matrix equations
- 46) Which method is used to solve linear algebraic equations?
- Newton-Raphson method
 - Bisection method
 - Gaussian elimination method
 - Simpson's Rule
- 47) Central difference formula is generally used when the point lies:
- At the beginning
 - At the end
 - Near the middle
 - Outside the table
- The error in trapezoidal rule decreases when:
- 48)
 - Step size increases
 - Step size decreases
 - Number of intervals decreases
 - Function becomes linear
- 49) Simpson's one-third rule is applicable when the number of subintervals nnn is:
- Odd
 - Even
 - Any number
 - Prime number
- 50) The order of accuracy (degree of precision) of Simpson's one-third rule is:
- First
 - Second
 - Third
 - Fourth
- 51) Compared to Simpson's one-third rule, Simpson's one-eighth rule is:

- A. Less accurate
 - B. More accurate
 - C. Equally accurate but used in different cases
 - D. Outdated
- 52) The degree of $y(x)$ in Trapezoidal rule is:
- A. One
 - B. Two
 - C. Three
 - D. Six
- 53) The degree of $y(x)$ in Simpson's 3/8 rule is:
- A. 1
 - B. 2
 - C. 3
 - D. 6
- 54) Trapezoidal formula is also known as:
- A. Simpson's Rules
 - B. Coordinate Method
 - C. Average end Area Method
 - D. None of the Above
- 55) Which of the following is not a type of numerical method?
- A. Root-finding method
 - B. Analytical method
 - C. Interpolation method
 - D. Integration method
- 56) What is the main purpose of numerical methods in solving mathematical problems?
- A. Approximate solution
 - B. Provide exact solution
 - C. Eliminate error
 - D. Simplify calculations
- 57) Gauss-Seidel method is preferred over Jacobi method because:
- A. It is simpler
 - B. It converges faster
 - C. It gives exact solution
 - D. It avoids iteration
- 58) Which of the following is a **direct method** for solving simultaneous algebraic equations?
- A. Jacobi's method
 - B. Gauss elimination
 - C. Relaxation method
 - D. Gauss-Seidel method
- 59) Numerical differentiation is based on:
- A. Difference formulas
 - B. Integration formulas
 - C. Matrix formulas
 - D. Elimination formulas
- 60) The condition for stability in numerical method is related to:
- A. Convergence
 - B. Accuracy
 - C. Efficiency
 - D. Both A and B
- 61) What is the primary disadvantage of the Bisection method?
- A. Slow convergence
 - B. Requires derivatives
 - C. Limited to linear equations
 - D. None of the above
- 62) **Numerical differentiation is used to find:**

- A. Integration
- B. Derivative
- C. Root
- D. Matrix

63) **Numerical integration is used to find:**

- A. Area under curve
- B. Derivative
- C. Roots
- D. Matrix

64) Simpson's 1/3 rule uses polynomial of degree:

- A. 1
- B. 2
- C. 3
- D. 4

65) Simpson's 3/8 rule is applicable when number of intervals is:

- A. Multiple of 2
- B. Multiple of 3
- C. Multiple of 4
- D. Any number

66) Gauss elimination method is used to solve:

- A. Linear equations
- B. Differential equations
- C. Integral equations
- D. Nonlinear equations

67) Gauss-Seidel method requires:

- A. Initial approximation
- B. Exact value
- C. Graphical solution
- D. Matrix inverse

68) In numerical integration, h represents:

- A. Error
- B. Step size
- C. Root
- D. Interval number

69) Simpson's 1/3 rule requires number of intervals to be:

- A. Odd
- B. Even
- C. Prime
- D. Zero

70) Numerical integration methods are used to evaluate:

- A. Definite integrals
- B. Indefinite integrals
- C. Differential equations
- D. Algebraic equations

71) The trapezoidal rule gives:

- A. Exact solution
- B. Approximate solution
- C. Analytical solution
- D. Graphical solution

72) Gauss-Seidel method improves solution by:

- A. Repetition
- B. Elimination

- C. Substitution
 - D. Integration
- 73) Numerical differentiation calculates:
- A. Slope
 - B. Area
 - C. Root
 - D. Matrix
- 74) Simpson's $3/8$ rule uses:
- A. 2 intervals
 - B. 3 intervals
 - C. 4 intervals
 - D. 5 intervals
- 75) Gauss elimination uses:
- A. Row operations
 - B. Column operations
 - C. Graph
 - D. Iteration
- 76) Numerical methods are particularly useful for:
- A. Complex equations
 - B. Simple equations
 - C. Linear functions
 - D. Constant functions
- 77) Trapezoidal rule assumes the function between two points is:
- A. Linear
 - B. Quadratic
 - C. Cubic
 - D. Constant
- 78) Simpson's $1/3$ rule is more accurate because it uses:
- A. Straight lines
 - B. Parabolic arcs
 - C. Circles
 - D. Cubic curves
- 79) Gauss-Seidel method converges when matrix is:
- A. Diagonally dominant
 - B. Singular
 - C. Zero
 - D. Rectangular
- 80) Numerical differentiation formulas depend on:
- A. Difference tables
 - B. Matrices
 - C. Graphs
 - D. Roots
- 81) Trapezoidal rule divides area into:
- A. Triangles
 - B. Trapeziums
 - C. Squares
 - D. Circles
- 82) Simpson's $1/3$ rule uses:
- A. 3 ordinates
 - B. 2 ordinates
 - C. 4 ordinates
 - D. 5 ordinates
- 83) Gauss elimination method transforms matrix to:
- A. Triangular form
 - B. Diagonal form

- C. Identity form
 - D. Square form
- 84) Numerical integration is also called:
- A. Quadrature
 - B. Interpolation
 - C. Extrapolation
 - D. Iteration
- 85) Gauss-Seidel method stops when:
- A. Desired accuracy achieved
 - B. Error increases
 - C. Matrix changes
 - D. Values decrease
- 86) Simpson's 3/8 rule fits:
- A. Straight line
 - B. Parabola
 - C. Cubic curve
 - D. Circle
- 87) Numerical differentiation requires:
- A. Equal intervals
 - B. Unequal intervals
 - C. Random values
 - D. Exact function
- 88) Gauss elimination is completed using:
- A. Back substitution
 - B. Forward substitution
 - C. Integration
 - D. Differentiation
- 89) Trapezoidal rule error depends on:
- A. Step size
 - B. Matrix size
 - C. Iterations
 - D. Roots
- 90) Simpson's 3/8 rule divides the interval into:
- A. Two equal parts
 - B. Three equal parts
 - C. Four equal parts
 - D. Five equal parts
- 91) Gauss-Seidel method is best suited for:
- A. Large systems
 - B. Small systems
 - C. Single equation
 - D. Differential equation
- 92) Numerical integration accuracy increases when:
- A. Step size decreases
 - B. Step size increases
 - C. Iterations increase
 - D. Roots decrease
- 93) Difference table is used in:
- A. Numerical differentiation
 - B. Gauss elimination
 - C. Matrix method
 - D. Graph method
- 94) Simpson's rules belong to:
- A. Numerical integration
 - B. Numerical differentiation

- C. Root finding
 - D. Matrix solving
- 95) Numerical methods are implemented using:
- A. Computers
 - B. Graphs
 - C. Tables
 - D. Calculators
- 96) Simpson's 3/8 rule is used when Simpson's 1/3 rule:
- A. Cannot be applied
 - B. Gives exact result
 - C. Is slower
 - D. Is complex
- 97) Gauss-Seidel method gives:
- A. Approximate solution
 - B. Exact solution
 - C. Analytical solution
 - D. Graphical solution
- 98) Numerical integration is also known as:
- A. Differentiation
 - B. Quadrature
 - C. Interpolation
 - D. Extrapolation
- 99) Numerical integration is used to find:
- A. Area under curve
 - B. Root
 - C. Matrix
 - D. Derivativ
- 100) Trapezoidal rule is based on:
- A. Linear interpolation
 - B. Quadratic interpolation
 - C. Cubic interpolation
 - D. Constant interpolation
- 101) The truncation error in trapezoidal rule is of order:
- A. h
 - B. h^2
 - C. h^3
 - D. h^4
- 102) The truncation error in Simpson's 1/3 rule is of order:
- A. h^2
 - B. h^3
 - C. h^4
 - D. h^5
- 103) The truncation error in Simpson's 3/8 rule is of order:
- A. h^2
 - B. h^3
 - C. h^4
 - D. h^5
- 104) The degree of precision of trapezoidal rule is:
- A. 0
 - B. 1

- C. 2
 - D. 3
- 105) The degree of precision of Simpson's $1/3$ rule is:
- A. 1
 - B. 2
 - C. 3
 - D. 4
- 106) The degree of precision of Simpson's $3/8$ rule is:
- A. 1
 - B. 2
 - C. 3
 - D. 4
- 107) Gauss elimination method fails when:
- A. Zero pivot occurs
 - B. Matrix is square
 - C. Matrix is diagonal
 - D. Matrix is symmetric
- 108) Gauss-Seidel method converges when matrix is:
- A. Diagonally dominant
 - B. Singular
 - C. Symmetric
 - D. Rectangular
- 109) Forward difference formula is most accurate when:
- A. Beginning of table
 - B. End of table
 - C. Middle of table
 - D. Outside table
- 110) Backward difference formula is best suited when value lies near:
- A. Beginning
 - B. End
 - C. Middle
 - D. Outside
- 111) Central difference formula provides:
- A. Minimum accuracy
 - B. Maximum accuracy
 - C. Same accuracy
 - D. No accuracy
- 112) The general quadratic formula method is used for:
- A. Numerical differentiation
 - B. Numerical integration
 - C. Root finding
 - D. Linear equations
- 113) Simpson's $1/3$ rule cannot be applied when:
- A. Intervals odd
 - B. Intervals even
 - C. Step size constant
 - D. Function continuous

114) The composite trapezoidal rule improves accuracy by:

- A. Increasing intervals
- B. Decreasing intervals
- C. Changing formula
- D. Changing function

115) Apply gauss siedel method to solve the equations; $20x+y-2z+10, 3x+20y-z=-18, 2x-3y+20z=25$ assume initial guess

$x_0 = y_0 = z_0 = 0$ then value of x after first iteration is _____

- A. 0.85
- B. 0.5
- C. 1.85
- D. 1.25